

## Topic 6: Chain-of-Thought Prompting

### 6.1 What is Chain-of-Thought (CoT)?

- Chain-of-thought prompting encourages the model to reason step by step before giving a final answer
- Dramatically improves accuracy on multi-step reasoning tasks: math, logic, planning, analysis
- The model's intermediate reasoning steps help it avoid jumping to wrong conclusions
- Zero-shot CoT: simply add 'Let's think step by step.' to your prompt
- Few-shot CoT: provide examples that include the reasoning steps, not just the answer

### 6.2 Zero-Shot Chain-of-Thought

- The phrase 'Let's think step by step' or 'Think through this carefully before answering' triggers CoT behaviour
- Works surprisingly well on arithmetic, logical deduction, and common-sense reasoning
- For final answer extraction add: 'After reasoning, state your final answer clearly on the last line.'

WITHOUT CoT (often wrong on tricky problems):

A bat and ball cost \$1.10. The bat costs \$1 more than the ball. How much does the ball cost?  
Answer: 10 cents [WRONG - common wrong answer]

WITH CoT:

A bat and ball cost \$1.10. The bat costs \$1 more than the ball.  
How much does the ball cost? Let's think step by step.

Step 1: Let ball = x cents

Step 2: bat = x + 100 cents

Step 3:  $x + (x + 100) = 110 \Rightarrow 2x = 10 \Rightarrow x = 5$

Final answer: The ball costs 5 cents.

### 6.3 Few-Shot Chain-of-Thought

- Provide 2-3 examples where both the reasoning steps AND the final answer are shown
- The model learns to apply the same reasoning pattern to new problems
- Most effective for: multi-step math, logical puzzles, code debugging, legal/medical reasoning

Q: Roger has 5 tennis balls. He buys 2 more cans of 3 balls each. How many does he have?  
A: Roger starts with 5. He buys 2 cans x 3 balls = 6 balls.  $5 + 6 = 11$ . Answer: 11

Q: The cafeteria had 23 apples. They used 20 for lunch and bought 6 more. How many now?  
A: Start: 23. Used:  $23 - 20 = 3$ . Bought:  $3 + 6 = 9$ . Answer: 9

Q: A store had 50 shirts. They sold 30% and then received a shipment of 10. How many now?  
A:

## 6.4 When to Use CoT

- Use CoT when: the task requires multiple logical steps, arithmetic, or causal reasoning
  - Skip CoT for: simple factual lookups, single-step tasks, or when speed matters more than accuracy
  - CoT increases output length and latency -- trade-off between accuracy and speed
  - For production systems: use CoT in the reasoning pass, then extract just the answer
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